

Metalloproteomics, metalloproteomes, and the annotation of metalloproteins

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Metalloproteomics includes approaches that address the expression of metalloproteins and their changes in biological time and space. Metalloproteomes are investigated by a combination of approaches. Experimental approaches include structural genomics, which provides insights into the architecture of metal-binding sites in metalloproteins and establishes ligand signatures from the types and spacings of the metal ligands in the protein sequence. Theoretical approaches employ these ligand signatures as templates for homology searches in sequence databases. In this way, the number of metalloproteins in the iron, copper, and zinc metalloproteomes in various phyla of life has been estimated. Yet, manganese metalloproteomes remain poorly defined. Metals have catalytic and structural functions in proteins. However, additional functions have evolved. Proteins that control metal homeostasis and proteins that are metal-regulated bind metal ions transiently and are generally not accounted for in estimates from bioinformatics. Thus, metalloproteomes are dynamic and likely to be larger than present estimates suggest. This account discusses the assignment of transition metals in metalloproteins and the ensuing issues facing analytical chemists and structural and computational biologists. Biological and chemical selectivities render metal selection by metalloproteins either more stringent or less stringent depending on the metal homeostatic system of the organism, the subcellular location of the protein, and environmental factors. Failure to recognize the principles of metal utilization has led to assigning the wrong metal in metalloproteins and has missed some of the regulatory functions of transition metal ions.

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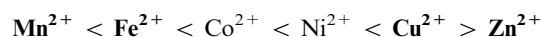
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1. Introduction

Systemic and cellular homeostatic control of transition metal ions is necessarily tight so as to provide essential amounts of metal ions while preventing toxicity in the context of available binding sites in metalloproteins, which are about 25% of all proteins. Many different steps lie between metal selection of an organism and metal selection of a metalloprotein (Fig. 1). These “selectivity filters” determine which metal ions enter the cell and reach its subcellular compartments. Eukarya have more selectivity filters than prokarya (Fig. 1). Coordination chemistry of ligands that handle the metal ions provides specificity during transit in the cytoplasm and transport through membranes. The metalloprotein provides only the last step in selecting the metal. How the correct metal ion is incorporated into a protein turns out to be a question of biology, in addition to the coordination chemistry of the metalloprotein.

This tutorial addresses the selection of the nutritionally essential metal ions from the first transition series for metalloprotein function. According to the Irving-Williams series¹ the divalent metal ions interact with ligands in the following order of affinity:



The Irving-Williams series holds for isolated proteins.² By way of exclusion in animals and humans, cobalt is utilized only when bound to the corrin cofactor in the form of vitamin B₁₂, a highly specialized handling device for this metal ion, and

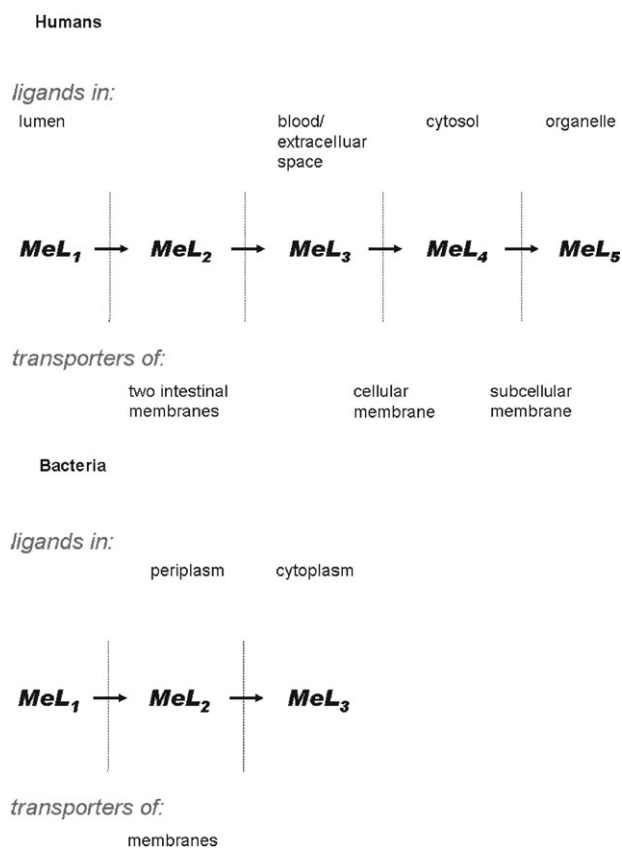


Fig. 1 Selectivity filters in humans and bacteria. The subscripts for the ligand (L) in metal–ligand complexes (MeL) indicate the type and not the number of the ligands. Taking into account the function of the liver in re-distribution of nutrients to human tissues, up to ten membranes may have to be crossed to reach a particular cell.¹¹⁴

nickel does not have an established molecular function, though there are health effects of nickel deficiency in animals.³ Prokarya may also use cobalt ions that are not corrin-bound in their enzymes.⁴ Nickel enzymes exist in plants and in prokarya, sometimes associated with the hydrocorphin cofactor F_{430} .⁵ Thus, for animals and humans, there is a sufficiently wide gap between those transition metal ions that bind rather weakly (Mn^{2+} and Fe^{2+}) and those that bind rather tightly (Cu^{2+} and Zn^{2+}), and the main issue is how proteins compete for these metals ions (shown in boldface above). Three of them are redox-active (Mn, Fe, Cu) and their affinities can be modulated by either oxidation (Mn, Fe) or reduction (Cu), while Zn is redox-inert in biological systems. When discussing the issue of metal selectivity, I shall not consider competition for either Mg^{2+} or Ca^{2+} . The chemical selectivity of proteins for these two metal ions, in relation to zinc, has been discussed in detail.⁶

The Irving-Williams series considers relative affinities. Affinity for a metal in a metalloprotein is high if the function of the protein requires keeping the metal ion bound. Accordingly, for catalytic and structural metal ions, the equilibrium between the metalloprotein (MeP) and the free metal ion (Me) and the protein (P) is far to the right:

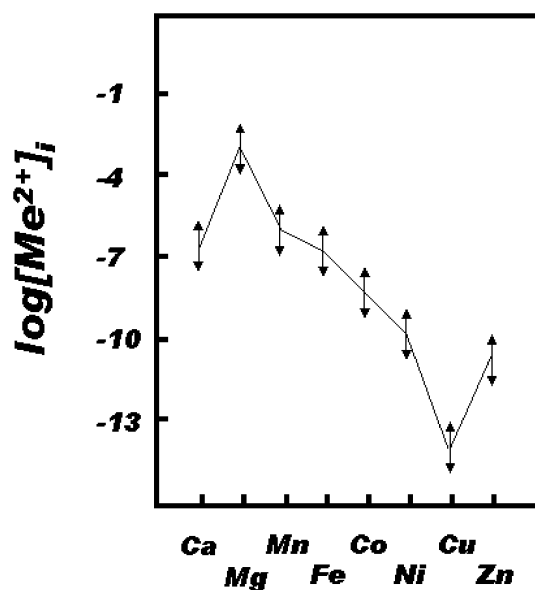
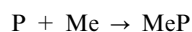
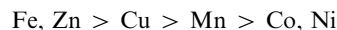


Fig. 2 Estimates of free divalent metal ion concentrations in a eukaryotic cell.¹²³ Sequestration of cobalt in the corrin ring, and presumably exclusion of nickel, leaves a sufficiently large gap of almost three orders of magnitude in concentrations between the weak binding (Mn, Fe) and the tight binding (Cu, Zn) transition metal ions.

If the metal were to dissociate from the protein it would compromise its function. When proteins function in the transport of metal ions, kinetic mechanisms affect dissociation of the metal ion. With knowledge about affinities of metalloproteins, the Irving-Williams series also predicts the cellular concentrations of free metal ions under equilibrium conditions (Fig. 2).

Determination of these free metal ion concentrations is important because they set the boundaries for possible regulatory functions and adverse effects on biological processes, such as competition with other metal ions.⁷ The relative affinities of the metal ions in the Irving-Williams series are also based on the same metal ion concentrations. However, total cellular concentrations differ for each metal ion and are an additional parameter to cope with in biological selectivity. In *Escherichia coli*, total magnesium is > 10 mM, calcium, zinc and iron are about $100 \mu M$, copper and manganese are about $10 \mu M$, and cobalt and nickel are even lower.⁸ In human organs, the metal concentrations follow the same trend (Table 1).⁹ Thus, the order of metal ion concentrations differs from the order of the metal ions in the Irving-Williams series:



An inescapable conclusion, based on equilibrium considerations, is that each metal ion must be buffered in a relatively narrow range in order not to interfere with the functions of the other metal ions. This physiological buffering is usually not considered when metal ions are added during expression, purification, or crystallization of a metalloprotein, with the consequence that a non-native metal can be incorporated into the metalloprotein.

Structural genomics has enhanced our knowledge about the roles of transition metal ions in biology in an unprecedented way. Characteristic types of ligands and spacings between the

Table 1 Metal concentrations in human organs in ppm ($\mu\text{g per g ash}$)⁹

	Liver	Kidney	Lung	Heart	Brain	Muscle
Mn	138	79	29	27	22	<4–40
Fe	16 769	7168	24 967	5530	4100	3500
Co	<2–13	<2	<2–8	—	<2	150 (?)
Ni	<5	<5–12	<5	<5	<5	<15
Cu	882	379	220	350	401	85–305
Zn	5543	5018	1470	2772	915	4688

ligands in the primary structure, so-called ligand signatures, determine the coordination environments of metals in proteins. These signatures have been used for homology searches in databases to predict putative metalloproteins.¹⁰ Adding information about the domain structure and the interactions of ligands with amino acids beyond the first coordination sphere made the prediction more robust and gleaned numerous new metal-binding sites from sequence and structure databases.^{11–13} In this way, the metalloproteomes of zinc, copper, and iron in different phyla were estimated from genome-wide studies.^{14–16} In the human genome, about 10% of the genes encode proteins with zinc sites, amounting to about 3000 zinc metalloproteins.¹⁴ The zinc proteome is likely to be even larger, since bioinformatics approaches do not consider signatures with non-sequential binding of ligands, zinc binding at protein interface sites, and permanent and transient zinc binding in sites with yet to be determined ligand signatures.^{17–19}

Significant efforts were spent on analyzing ligand preferences and geometries of metals in proteins, but determining the correct metal in a metalloprotein continues to be a challenge that is not readily met without additional information about the purification of the protein and the biology and the environment of the organism from which the protein was isolated. Isolated proteins may not contain a metal at all or experimental conditions may favor binding of a metal ion other than the native one.

2. Characterization of metalloproteins

The question concerning which metal ion a metalloprotein selects sometimes has no clear-cut answer, especially when the metalloprotein under investigation is from an organism with fewer “selectivity filters”. Traditionally, isolation and analytical characterization of the protein has established the metal requirement of a metalloprotein. An operational distinction was made between metalloproteins, which bind metal ions so tightly that the metal does not dissociate during purification, and metal–protein complexes, which may lose their metal during purification.²⁰ However, recently, metal sites in proteins have been predicted “*in silico*” based on sequence homologies. In this case, conclusions about affinities are tenuous and metalloproteins cannot be readily distinguished from metal–protein complexes. For instance, it is commonly assumed that all proteins classified as zinc-binding indeed bind zinc and not any other metal ion, an assumption that can be wrong. Thus, “zinc finger domain” became a generic term in the analysis of protein structures without actually determining the zinc content of the protein, in most instances. On the other

hand, the term “metalloprotease” does not specify the type of metal, though it usually implies that zinc is the metal. In this article, I will show that the two approaches, the isolation of metalloproteins from a tissue, cell or an expression system, and the prediction of metalloprotein sites by bioinformatics, provide ample possibilities for incorrectly identifying metal ions in metalloproteins. Prediction of metal sites is not an end point but a starting point for experimental verification. High resolution structural studies often ignore the specifics of the metal metabolism of the organism from which the protein originates. As a consequence, in structural genomics there is considerable ambiguity in the assignment of some metals in metalloproteins.

2.1 Isolation of a metalloprotein from a tissue or from cells

If the metal ion binds tightly to a protein and is not lost during purification, the nature of the metalloprotein can be established by direct analysis. In most cases of metalloproteins containing a single metal ion, only one type of metal ion is found. Purification of a putative metalloprotein can result in the preparation of a pure protein that does not contain a metal ion, however. One common cause is the use of chelating agents, such as EDTA, and/or reducing agents, such as DTT, both of which form very strong metal complexes,²¹ during isolation of the protein. Therefore, the Protein Data Bank (pdb.org) contains many structures of metalloproteins in their apoforms or with an incorrectly assigned metal ion.^{22,23} In further functional studies of purified proteins, an activating metal usually reveals its presence, but any inhibitory metal ion needs to be removed. Protein tyrosine phosphatases (PTPs) illustrate this point. They are not considered to be zinc proteins. However, zinc inhibits human PTP-1B with a K_i value of 15 nM.²⁴ The commercial enzyme (Calbiochem) is stored in 1 mM EDTA and 1 mM dithiothreitol (DTT). Under these conditions, metal inhibition is not apparent.

When cells are grown in the laboratory, many “selectivity filters” (Fig. 1) are absent. Culture conditions differ from native conditions. For example, growing yeast in a cobalt(II)-enriched medium results in the incorporation of cobalt instead of zinc into alcohol dehydrogenase.²⁵ Also, some coordination environments with tightly bound metal ions can exchange metal ions rapidly, resulting in the incorporation of metal ions that may be present in the buffer. Therefore, analysts should be aware of the procedures to prevent metal contamination in their experiments.²⁶ Metal ions, such as iron and zinc, are ubiquitous and thus may be present in much higher quantities in the laboratory environment than the “selectivity” filters allow them to be present *in vivo*.

Metal ions in metalloproteins can be replaced *in vitro*, demonstrating that metalloproteins are usually not highly selective of metal ions, and that other metal ions, even toxic ones, can bind and affect function.^{27–29} In some cases, there is less enzymatic activity with a non-native metal, but hyperactivation with a non-native metal ion, *e.g.*, copper instead of zinc in leucine aminopeptidase, also occurs.³⁰ Thus, maximal activity is not a sufficient criterion for proof of the correct metal. Metal ions that are largely excluded by animals and humans, *i.e.*, non-corrin cobalt or nickel, fulfill functions

relatively well *in vitro* in many metalloproteins. For this reason, they can be used as spectroscopic probes to investigate metal sites in metalloproteins.³¹ Metal ions that activate an apoprotein within a relatively narrow range of concentrations, usually at stoichiometric amounts, may inhibit when in excess. Thus, the true function of the metal may be revealed only if experiments are performed in an appropriate range of metal ion concentrations.

2.2 Isolation of a recombinant metalloprotein from an expression system

Preparation of a metalloprotein from a heterologous expression system can lead to the incorporation of metal ions other than the native one, especially when the metal ion in question is limiting in the growth medium. Expression of *Pseudomonas aeruginosa* azurin in *Escherichia coli* results in the production of the zinc protein, in addition to the native copper protein.³² Likewise, in expressed LIM proteins (acronym derived from the genes *lin-11*, *isl-1*, and *mec-3*), iron can be incorporated into what is believed to be a zinc-binding domain *in vivo*.³³ A common, but potentially misleading procedure involves enriching the growth medium with the metal ion that is assumed to be the correct one based on sequence homologies. Without additional evidence about the nature of the native metal ion, metal assignment in expressed proteins should be approached with caution, because both the metal regulatory system of the host and the environment for supporting growth can affect the results. One wonders how many expressed metalloproteins contain a non-native metal.

2.3 Metal content depending on species

The observation that metalloenzymes can employ different metals *in vitro* supports the notion that the protein is only the last, and perhaps the least important, step in selecting the metal.

Carbonic anhydrase (CA) was the first zinc protein discovered.³⁴ CAs of α -, β -, and γ -classes have significantly different primary sequences. α - and γ -classes have three histidyl ligands, while the β -class binds the metal with one histidine and two cysteines. In marine diatoms, two additional classes, δ and ζ , exist. The δ -CA has an active site similar to that of the α -class CAs while the ligands of the metal in the ζ -CA are the same as in the enzymes of the β -class. The ζ -CA can be a cadmium enzyme. In contrast to mammalian CA, metal exchange in the active site of diatom CA is fast, suggesting that at the low metal ion concentrations in the marine environment, the organism can use cadmium when zinc is rare.³⁵ When overexpressed in *Methanosarcina acetivorans*, the archaeal γ -class CA from *Methanosarcina thermophila* is an iron enzyme.³⁶ Based on the Irving-Williams series, iron binding is at least one order of magnitude weaker than zinc binding. Purification under aerobic conditions results in a loss of iron as Fe^{3+} and incorporation of Zn^{2+} .³⁷ The authors caution against assigning metalloenzymes from anaerobic species, when overexpressed in heterologous systems, as zinc enzymes.³⁶ Human CA VIII, in contrast, does not bind zinc at all. It is catalytically silenced by replacing

one of the essential histidine ligands with an arginine and restricting access of carbon dioxide to the active site by bulky side chains. The electron density in the active site was modeled as a chloride ion.³⁸

Alcohol dehydrogenase (ADH), another classic zinc enzyme, is an iron enzyme in yeast mitochondria. Bacterial species, such as *Zymomonas mobilis*, also have an iron-dependent ADH.³⁹ ADHs with a second, non-catalytic zinc site with four cysteines as ligands reveal a relationship to iron metabolism. The coordination in this site resembles that of iron in rubredoxin. Furthermore, the chirality of the protein fold around this site resembles that of iron in bacterial ferredoxins.⁴⁰ The crystal structure of L-threonine dehydrogenase from *Pyrococcus horikoshii* revealed a protein fold similar to other ADHs.⁴¹ However, while the non-catalytic zinc ion is present, the catalytic zinc ion is missing in the structure, presumably because it had been lost during preparation of the enzyme.

The cytosolic and the plasma superoxide dismutases (SOD1 and 3) have binuclear copper/zinc sites. The mitochondrial enzyme (SOD2), however, has a mononuclear manganese site. Iron is the cofactor in bacterial SODs. There are also so-called cambialistic SODs that can use either manganese or iron. Other bacteria, such as *Streptomyces coelicolor*, have a nickel SOD.⁴² In yeast mitochondria, where iron concentrations are two orders of magnitude higher than manganese concentrations, iron can be incorporated into SOD2 only when the cells are starved for manganese, or mitochondrial iron homeostasis is interrupted.⁴³

Human glyoxalase I is a zinc enzyme that belongs to a structural family that includes an iron-dependent enzyme.⁴⁴ However, the *Escherichia coli* enzyme is a nickel enzyme.⁴⁵ Even though the primary structures are homologous and the 3D structures similar, the *Escherichia coli* enzyme accommodates zinc in a trigonal bipyramidal, pentacoordinate environment and is inactive. An octahedral, hexacoordinate environment, as found in the nickel enzyme, is thought to be a prerequisite for activity. Human glyoxalase II was believed to also be a zinc enzyme, but containing a binuclear zinc site.⁴⁶ However, recent work suggests it is active when only one zinc ion is bound. For the originally reported stoichiometry of 1.5 Zn and 0.7 Fe, iron was considered a contaminant,⁴⁶ but given the availability of zinc and iron, human glyoxalase II is likely an Fe, Zn enzyme.⁴⁷

2.4 Metal content depending on location

Which metal ion a metalloprotein acquires also depends on its subcellular location. In bacteria, metal selectivity is related to where the protein folds.⁴⁸ Thus, in the cyanobacterium *Synechocystis* PCC 6803, the most abundant copper and manganese proteins in the periplasm are the cupins CusA and MncA. Both proteins have identical coordination environments with three histidine ligands and one glutamate ligand. Hence, the proteins do not provide any selectivity. The manganese protein folds in the cytosol where the manganese (II) concentrations are several orders of magnitude higher than those of zinc and copper. Once bound, manganese is kinetically trapped in the protein. The copper protein, however, folds in the periplasm. With the strategem of folding

the protein in different compartments and using two different export pathways, the metal-binding preference dictated by the Irving-Williams series is overruled. Another case where location is important is the compartmentation of eukaryotic cells. In addition to glyoxalase II and ADH, other mitochondrial proteins bind iron instead of zinc. Because of the extensive handling of iron in mitochondria for Fe/S cluster synthesis and heme synthesis, discrimination between zinc and iron is a specific issue in this organelle. Both human Mia40 (mitochondrial intermembrane space import and assembly) and human mitoNEET (where the acronym NEET stands for the amino acid sequence Asn-Glu-Glu-Thr) were thought to be zinc proteins, but actually bind iron.^{49,50} Likewise, the endoplasmic reticulum protein Miner1 has a domain that was annotated as a zinc finger but turns out to have a fold similar to mitoNEET and to contain a 2Fe/2S cluster.⁵¹ Glutaredoxin-2 is a dimer that is bridged by an iron-sulfur cluster.⁵² The 2Fe/2S cluster-bridge includes the sulfur ligands from the *N*-terminal, active-site cysteines of each monomer and two reduced glutathione molecules.⁵³ The dimer is inactive and redox-regulated in such a way that monomerization generates activity.⁵⁴ The presence of iron at a protein interface site is noteworthy because usually zinc ions are utilized for bridging proteins.⁵⁵

Uncertainty persists in the assignment of metals in the two enzymes of the *N*-terminal methionine excision pathway, formyl peptidase (peptide deformylase) and methionine aminopeptidase. *Escherichia coli* formyl peptidase was initially thought to be a nickel enzyme, but was then suggested to be an iron enzyme.^{56,57} In bacteria that virtually eliminated the need for iron, such as *Borrelia burgdorferi*, formyl peptidase is a zinc enzyme.⁵⁸ Yeast methionine aminopeptidase (MetAP) was thought to be a cobalt enzyme because zinc was found to be inhibitory. However, considering that zinc concentrations are at least three orders of magnitude higher than cobalt concentrations, MetAP I is most certainly a zinc enzyme and no cytoplasmic enzyme utilizes cobalt in yeast.⁵⁹ *Escherichia coli* MetAP is also thought to be an iron enzyme.^{60,61} The two enzymes of this pathway have been detected in human mitochondria.⁶² The structure of the human mitochondrial formyl peptidase was solved with cobalt in the active site.⁶³ However, the mitochondrial enzyme in plants (*Arabidopsis thaliana*) is a zinc enzyme.⁶⁴ For the human mitochondrial MetAP1D, cobalt is also the best activator.⁶⁵ However, as pointed out above, optimal enzymatic activity is not decisive when determining the correct metal. It will be interesting to learn whether these human mitochondrial enzymes are iron or zinc enzymes, and more generally, to which extent iron is utilized instead of zinc in what are believed to be zinc proteins in mitochondria.

Though investigators may learn a lot about the structure and function of these enzymes with an enzymatically active metal, clearly more consideration needs to be given to the physiological metal coordination. Geometries of the metal-ion binding sites in protein structures need to be restrained properly for identifying the metal ion in a metalloprotein from crystallographic data.²³ Yet, this procedure does not guarantee that the metal identified is the one that the protein binds *in vivo*.

2.5 Metal content depending on metal ion availability and redox state

Most of the above discussion focused on enzymes where specific metal ion chemistries are exploited for catalysis. Structural sites do not rely on the catalytic prowess of the metal, and they utilize the redox-inert zinc (or calcium) ion. But which metal ions are employed in proteins with other functions in metal metabolism? Proteins that bind metals transiently might not have such stringent requirements. Considerable uncertainties exist in the assignment of metal ions in other types of proteins, such as transporters and sensors. Thus, the zinc transporter Zip14 (zrt, irt-like protein) actually transports iron, and the zinc transporter Zip8 transports manganese and cadmium.^{66–68} However, metal sensors are believed to be specific for a particular metal ion.⁶⁹ Apparently, the issue is not so much one of selectivity of the proteins, which bind different metal ions *in vitro*, but rather one of their responsiveness to metal ion concentrations *in vivo*.⁷⁰ The metalloregulators of the Fur family continue to pose significant challenges in protein structure determinations with regard to clarifying how individual members can be sensors for iron (Fur), zinc (Zur), manganese (Mur), nickel (Nur), or for metabolites, such as heme (Irr) or peroxide (Per). *Escherichia coli* Fur was identified as a zinc metalloprotein.⁷¹ Because it is difficult to populate the different metal-binding sites with the correct metal ions in the expressed proteins, crystal structures were determined on proteins with zinc ions. The original interpretation from the structure of *Pseudomonas aeruginosa* Fur, namely that zinc(1) is the regulatory site occupied by iron *in vivo* and that zinc(2) is a structural site, has now been reversed based on theoretical approaches and the structure of *Vibrio cholerae* Fur.^{72,73} *Mycobacterium tuberculosis* FurB and *Bacillus subtilis* PerR contain a zinc ion in an additional site with four cysteine ligands.^{74,75} Interpretation of metal binding in these two proteins hinges on the redox state of these cysteines and the availability of zinc ions.

Metal ions bind to proteins transiently when the metal regulates protein function or the protein regulates metal metabolism.⁷⁶ Mammalian metallothionein (MT), a protein that binds seven divalent metal ions, or an even higher number of monovalent metal ions, with the sulfur ligands from twenty cysteines will serve as an example for such a dynamic situation.⁷⁷ Purification of MTs resulted in species that had different metal ions, with binding that depended on the type of tissue and the exposure of the organism to metals (Zn, Cu, Cd). For further studies, the isolated protein was brought into a state that contained only one metal, thus neglecting the fact that variable metal composition is an inherent property of the protein and reflects its natural environment. Isolation of MTs also needs to consider the cellular and subcellular location of the protein and the cellular redox state.⁷⁸ MTs are not only cytosolic, they can also reside in the nucleus and in the intermembrane space of mitochondria. In addition, MTs are exported from cells and are taken up by cells through a receptor-mediated mechanism, in which the protein remains in an endocytotic compartment and the metal is translocated to the cytosol.⁷⁹ For analytical characterization of the MT protein, controlling the redox state and the concentrations of

both metal ions and the protein is critical. The presence of chelating or redox agents, or simply dilution of the protein, changes its metalation. Owing to different binding constants for the seven bound zinc ions, MT exists in states with different metal loads, Zn₇T (“MT”), Zn₆T, Zn₅T, and Zn₄T, depending on the total concentration of the protein and the availability of zinc ions.⁸⁰ Given the low affinity for the seventh zinc ion and only picomolar cellular free zinc ion concentrations, Zn₇T cannot exist under normal physiological conditions. In addition, MT exists in oxidized forms, polymerized forms, thiol-modified forms, and mixed-metal species under specific environmental exposures.^{81,82} The thiolate coordination environment of the metals confers kinetic lability so that MT can readily swap metals with other proteins. Zinc MT can take up copper that is bound to the amyloid- β peptide, A β _{1–40} and it can collect cadmium bound to transcription factors with canonical zinc finger (ZnN₂S₂) motifs.^{83,84} In both cases, the resulting MT has a mixed-metal composition. Since the metal ion determines the protein conformation of MT and zinc finger domains, changing metal loads alters the conformational landscape of these proteins, which may be important for yet unknown interactions and functions.⁸⁵

An area of interest is the redox regulation of metal sites in proteins. In the case of the redox-inert zinc ion, redox reactions are possible and involve the ligand instead of the central atom.⁸⁶ Redox-zinc switches include redox activity of the sulfur (cysteine) ligands and concomitant dissociation/association of zinc, thus coupling regulation of protein structure and function to redox metabolism, converting redox signals into zinc signals, and establishing redox control of zinc functions.⁸⁷ Ligand modification and zinc release can be investigated with spectrophotometric methods.⁸⁸ Screening for redox-active cysteine pairs in the Protein Data Bank retrieved a total of 73 pairs (in 53 proteins) that are associated with expulsion of a metal ion, such as zinc.⁸⁹ In their oxidized states, these metalloproteins do not contain a metal ion. Thus, in structural studies, the redox state of such cysteines is critical for establishing whether or not the protein binds a metal ion.

The discussion about transient metal sites makes it clear that the metal may not be present in a metalloprotein or a site may not be fully occupied. Analyses then will show substoichiometric amounts of the metal in a protein, which is an inherent reason why stoichiometries can have non-integral numbers. Non-integral numbers also can result from metal ion binding between subunits. Determination of metal stoichiometries requires highly accurate analytical data. Obtaining such data is often limited by obtaining sufficiently pure protein and accurately measuring protein concentrations. If metals bridge proteins, not only subunit molecular masses but also the quaternary structure need to be determined.

2.6 Binuclear metal sites in metalloproteins

In some metalloproteins, metal ions are in a clustered arrangement. Their proximity can be examined with a variety of spectroscopic methods.⁹⁰ There are homometallic and heterometallic binuclear metal sites (Table 2).^{91–98} The latter make the issue of putting the correct metal into the correct site a challenging task. How do these proteins obtain different

metals exactly at a 1 : 1 ratio when metal ion concentrations are as different as in the case of copper and zinc? For Cu, Zn SOD, a copper metallochaperone, CCS (copper chaperone for SOD) inserts copper. It is not known whether the source of zinc is a pool of zinc ions or a zinc protein.

Most of the binuclear zinc metalloproteinases function with one zinc ion, but the type of metal and the presence of a second metal affect the function of the proteins.^{99,100} Therefore, the second metal has been termed co-catalytic.¹⁰¹ Prokaryotic proteins can be promiscuous regarding their metal selection. In the periplasm, where metal control is poor, β -lactamase is a zinc (Zn, Zn) enzyme, whereas in the cytoplasm, where control of zinc is tight, it is a Zn, Fe enzyme.¹⁰² The selection of different metal ions in the co-catalytic site of metalloenzymes with binuclear sites could be a major theme in enzymatic regulation, in both activation and inhibition, and it may determine substrate specificity.¹⁰³ A weak zinc-binding site will not be occupied by zinc but by another metal ion, such as iron, if it is available at higher concentrations. Thus, a distinction should be made between metalloproteins with genuine binuclear sites, where both metal ions are present, and those with conditional binuclear sites, where the second metal binds transiently and presumably has a regulatory function. A co-catalytic metal that is lost during purification can not be identified readily. The identity could be addressed, though, if the availability of metal ions in the organism were considered, and if binding constants of the metal ions were measured.

There are metalloenzymes, in which a second metal binds in proximity to the catalytic metal and inhibits enzymatic activity. Zinc inhibits carboxypeptidase A, a zinc proteinase, with a K_i of 0.5 μ M.¹⁰⁴ Chemical and physical characterization of the interaction suggested binding of the inhibitory zinc to glutamate-270, a hydroxide that bridges the catalytic and inhibitory zinc, and a chloride ion.¹⁰⁵ The suggestion proved to be correct when the crystal structure of the zinc-inhibited protein was solved.^{106,107} Other zinc enzymes may use the same mechanism of zinc inhibition.¹⁰⁸

Many enzymes in phosphate ester hydrolysis are metallo-enzymes. Alkaline phosphatase binds two zinc ions in a trinuclear Zn, Mg site. Calcineurin and purple acid phosphatases are binuclear Zn, Fe enzymes. Employing iron generates a new activity, namely redox modulation of enzymatic activity by the second metal ion. For calcineurin, the Zn, Fe³⁺ (ferric)

Table 2 Examples of binuclear metal sites in metalloproteins

Metals	Protein (species)	Ref.
<i>Homometallic</i>		
Mn–Mn	Arginase (rat liver)	91
Fe–Fe	Acyl-ACP-desaturase (<i>Ricinus communis</i>) ^a	92
Ni–Ni	Urease (<i>Klebsiella aerogenes</i>)	93
Cu–Cu	Hemocyanin (<i>Panulirus interruptus</i>)	94
Zn–Zn	Leucine aminopeptidase (bovine lens)	95
<i>Heterometallic</i> ^b		
Ni–Fe	Hydrogenase (<i>Desulfomicrobium baculatum</i>)	96
Zn–Fe	Calcineurin (human)	97
Zn–Cu	Superoxide dismutase (bovine erythrocytes)	98

^a also many 2Fe/2S cluster proteins. ^b not included are binuclear sites containing a transition metal ion and magnesium or calcium.

enzyme is active while the Zn, Fe²⁺ (ferrous) enzyme is inactive. Redox-regulation of these enzymes could extend to the di-iron state. Evidence for an Fe²⁺, Fe³⁺ enzyme that may be activated by either reduction or oxidation was provided.^{109,110} It is not known whether the combination of metal ions (Zn, Mn, Mg) observed in the active sites of other phosphatases, such as some Ser, Thr phosphatases, reflects the usage of the native metal ions. Given the low concentrations of manganese in human tissues (Table 1), the usage of manganese in enzymes requires scrutiny.

The human genome encodes 21 phosphodiesterases (PDEs). The crystal structures of several human PDEs have been reported.¹¹¹ Their active sites contain zinc in combination with another metal ion, which is believed to be magnesium, but it can be Mn²⁺ *in vitro*. Zinc binding to the co-catalytic site can be inhibitory. Thus, PDEs are likely Zn, Mg enzymes.

The presence of manganese (II) in the structures of metalloproteins needs to be reconciled with its usage *in vivo*. After a critical examination of the magnesium requirement for enzymatic hydrolysis of phosphate esters, it was noted that the physiological significance of magnesium is often neglected when manganese can serve as a substitute.¹¹² Accordingly, the magnitude of the binding constant and the availability of the metal ion in the cellular environment must be considered when discussing the selection of the correct metal and when distinguishing between 2-metal ion and 1-metal ion catalytic mechanisms.

3. Conclusions and prospects

I used selected examples to illustrate that (i) subcellular compartmentation of metal metabolism in eukarya affects metal availability for metalloenzymes; (ii) the same types of sites in proteins may bind different metal ions in different organisms or in different compartments; (iii) differences in homeostatic mechanisms and the ecology of the organism dictate metal usage; (iv) knowledge about metal availability, metal metabolism and its control, and metal affinity is necessary to discuss the metal requirement of a metalloenzyme, and to distinguish metalloproteins from metal–protein complexes; (v) prokaryotic metalloproteins can be promiscuous in their metal selection. Finally, on more speculative notes, metal switching may be physiologically significant, and fluctuating concentrations of metal ions may be important for biological regulation.

The following principles seem to minimize the competition between the essential metal ions of the first transition series. Current views hold that metallochaperones specifically deliver some metals to their target proteins via protein-protein interactions. Copper metallochaperones are readily affordable since the number of copper proteins is relatively small. They keep copper away from zinc sites.¹¹³ Thus, by keeping copper under “non-equilibrium selection”,¹¹⁴ iron and zinc remain the main contenders. In part, control is achieved by maintaining free zinc ions at very low concentrations, so that zinc does not interfere with the weaker binding sites where iron binds.¹¹⁴ In bacteria, free zinc ion concentrations are believed to be femtomolar, while in mammalian cells they are in the picomolar range.^{115–118} At such low concentrations, it is not

possible to populate low affinity sites with zinc. Also, the kinetics of zinc binding would be very slow.¹¹⁹ Using a k_{on} rate of 0.1 $\mu\text{M}^{-1}\text{s}^{-1}$ for human carbonic anhydrase II¹²⁰ and a free zinc ion concentration of 1 nM,⁷ the half-time for reconstituting the enzyme ($t_{1/2} = \ln 2/k[\text{Zn}]_{\text{free}}$) will be close to two hours, and thus, rather slow on a biological time scale. MT may solve this conundrum because it can transfer zinc to apoproteins or accept zinc from some zinc proteins *in vitro*.²⁴ Akin to the function of calmodulin in elevating the calcium concentrations from micromolar free calcium to millimolar bound calcium,¹¹⁴ MT provides zinc by carrier diffusion at concentrations that are at least three orders of magnitude higher than the concentrations of free zinc ions. In addition, iron protein synthesis is, in large part, controlled in mitochondria, where ferrochelatase inserts iron into protoporphyrin IX to form heme and where iron/sulfur cluster (ISC) assembly takes place.¹²¹ Evidence for a cytosolic iron metallochaperone that delivers iron to ferritin has been presented.¹²² Changing concentrations of metal ligands affords another level of control.¹¹⁴

In conclusion, metal selection of metalloproteins is an example where data-driven ‘omics’ approaches need to consider the complexity of biology. It cannot be addressed solely by purification of the protein and characterization of its metal binding, nor can it be predicted with high accuracy. Due to the large number of available structures, prediction can be reasonably accurate for catalytic and structural sites, where the metal ion usually binds permanently, but the scarcity of data for transient binding sites precludes using predictions. Metal selection needs to consider metal availability, the metal insertion machinery of the particular organism, and the biology of the cell. It is a biologically controlled process, in addition to the kinetic and thermodynamic control at the protein level. Last but not least, the relatively large number of structures of metalloproteins with non-native metals impedes discovery of drugs where metal binding is part of the mechanism of action.

Homeostatic control maintains a balance between supply and demand of metal ions. However, spill-over can occur if metal ion concentrations are perturbed or higher than normal concentrations of essential or toxic metal ions become available. A non-native metal ion can then substitute for the native metal ion. Metal substitution is an important mechanism in pathophysiology and toxicology, because functions of metalloproteins depend on the type of metal ion. A full understanding of the role of metal substitution in physiological control remains an issue for future investigations.

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References

- 1 H. Irving and R. J. P. Williams, *Nature*, 1948, **162**, 746–747.
- 2 K. A. McCall and C. A. Fierke, *Biochemistry*, 2004, **43**, 3979–3986.
- 3 F. H. Nielsen, in *Trace Elements in Human and Animal Nutrition*, ed. W. Mertz, Academic, San Diego, 5th edn, 1987, vol. 1, ch. 8, pp. 245–273.
- 4 M. Kobayashi and S. Shimizu, *Eur. J. Biochem.*, 1999, **261**, 1–9.
- 5 S. W. Ragsdale, *J. Biol. Chem.*, 2009, **284**, 18571–18575.
- 6 T. Dudev and C. Lim, *Chem. Rev.*, 2003, **103**, 773–787.
- 7 Y. Li and W. Maret, *Exp. Cell Res.*, 2009, **315**, 2463–2470.
- 8 L. A. Finney and T. V. O'Halloran, *Science*, 2003, **300**, 931–936.
- 9 G. V. Iyengar, W. E. Kollmer and H. J. M. Bowen, *The Elemental Composition of Human Tissues and Body Fluids*, Verlag Chemie, Weinheim, 1978.
- 10 B. L. Vallee and D. S. Auld, *Biochemistry*, 1990, **29**, 5647–5659.
- 11 J. C. Ebert and R. B. Altman, *Protein Sci.*, 2008, **17**, 54–65.
- 12 K. Goyal and S. C. Mande, *Proteins: Struct., Funct., Bioinf.*, 2008, **70**, 1206–1218.
- 13 R. Levy, M. Edelman and V. Sobolev, *Proteins: Struct., Funct., Bioinf.*, 2009, **76**, 365–374.
- 14 C. Andreini, L. Banci, I. Bertini and A. Rosato, *J. Proteome Res.*, 2006, **5**, 3173–3178.
- 15 C. Andreini, L. Banci, I. Bertini, S. Elmi and A. Rosato, *Proteins: Struct., Funct., Bioinf.*, 2007, **67**, 317–324.
- 16 C. Andreini, L. Banci, I. Bertini and A. Rosato, *J. Proteome Res.*, 2008, **7**, 209–216.
- 17 W. Maret, *J. Anal. At. Spectrom.*, 2004, **19**, 15–19.
- 18 W. Maret, *J. Trace Elem. Med. Biol.*, 2005, **19**, 7–12.
- 19 W. Maret, *Pure Appl. Chem.*, 2008, **80**, 2679–2687.
- 20 B. L. Vallee and W. E. C. Wacker, *The Proteins*, Academic Press, New York, 2nd edn, 1970, vol. 5.
- 21 A. Krezel, W. Lesniak, M. Jezowska-Bojczuk, P. Mlynarz, J. Brasun, H. Kozlowski and W. Bal, *J. Inorg. Biochem.*, 2001, **84**, 77–88.
- 22 M. Babor, S. Gerzon, B. Raveh, V. Sobolev and M. Edelman, *Proteins: Struct., Funct., Bioinf.*, 2008, **70**, 208–217.
- 23 H. Zheng, M. Chruszcz, P. Lasota, L. Lebiada and W. Minor, *J. Inorg. Biochem.*, 2008, **102**, 1765–1776.
- 24 A. Krezel and W. Maret, *JBIC, J. Biol. Inorg. Chem.*, 2008, **13**, 401–409.
- 25 A. Curdel and M. Iwatsubo, *FEBS Lett.*, 1968, **1**, 133–136.
- 26 M. Zief and J. W. Mitchell, *Contamination Control in Trace Element Analysis*, Wiley, New York, 1976.
- 27 B. L. Vallee, J. A. Rupley, T. L. Coombs and H. Neurath, *J. Am. Chem. Soc.*, 1958, **80**, 4750–4751.
- 28 W. Maret, in *Zinc Enzymes*, ed. I. Bertini, C. Luchinat, W. Maret and M. Zeppezauer, Birkhäuser, Basel, 1986, ch. 2, pp. 17–26.
- 29 W. Maret and M. Zeppezauer, *Methods Enzymol.*, 1988, **158**, 79–94.
- 30 J. M. Prescott, F. W. Wagner, B. Holmquist and B. L. Vallee, *Biochemistry*, 1985, **24**, 5350–5356.
- 31 W. Maret and B. L. Vallee, *Methods Enzymol.*, 1993, **226**, 52–71.
- 32 H. Nar, R. Huber, A. Messerschmidt, A. C. Fillipou, M. Barth, M. Jacquino, M. van de Kamp and G. W. Canters, *Eur. J. Biochem.*, 1992, **205**, 1123–1129.
- 33 P. M. Li, J. Reichert, G. Frey, H. R. Horvitz and C. T. Walsh, *Proc. Natl. Acad. Sci. U. S. A.*, 1991, **88**, 9210–9213.
- 34 D. Keilin and T. Mann, *Nature*, 1939, **144**, 442–443.
- 35 Y. Xu, L. Feng, P. D. Jeffrey, Y. Shi and F. M. M. Morel, *Nature*, 2008, **452**, 56–62.
- 36 S. R. MacAuley, S. A. Zimmermann, E. E. Apolinario, C. Evilia, Y.-M. Hou, J. G. Ferry and K. R. Sowers, *Biochemistry*, 2009, **48**, 817–819.
- 37 B. C. Tripp, C. B. Bell, F. Cruz, C. Krebs and J. G. Ferry, *J. Biol. Chem.*, 2003, **279**, 6683–6687.
- 38 S. S. Picaud, J. R. C. Muniz, A. Kramm, E. W. Pilka, G. Kochan, U. Oppermann and W. W. Yue, *Proteins: Struct., Funct., Bioinf.*, 2009, **76**, 507–511.
- 39 R. K. Scopes, *FEBS Lett.*, 1983, **156**, 303–306.
- 40 H. Eklund, B. Nordström, E. Zeppezauer, G. Söderlund, I. Ohlsson, R. Boiwe, B.-O. Söderberg, O. Tapia and C.-I. Brändén, *J. Mol. Biol.*, 1976, **102**, 27–59.
- 41 K. Ishikawa, N. Higashi, T. Nakamura, T. Matsuura and A. Nakagawa, *J. Mol. Biol.*, 2007, **366**, 857–867.
- 42 D. P. Barondeau, C. J. Kassmann, C. K. Bruns, J. A. Tainer and E. D. Getzoff, *Biochemistry*, 2004, **43**, 8038–8047.
- 43 A. Naranuntarat, L. T. Jensen, S. Pazicni, J. E. Penner-Hahn and V. C. Culotta, *J. Biol. Chem.*, 2009, **284**, 22633–22640.
- 44 A. D. Cameron, B. Olin, M. Ridderstrom, B. Mannervik and T. A. Jones, *EMBO J.*, 1997, **16**, 3386–3395.
- 45 M. M. He, S. L. Clugston, J. F. Honek and B. W. Matthews, *Biochemistry*, 2000, **39**, 8719–8727.
- 46 A. D. Cameron, M. Ridderstrom, B. Olin and B. Mannervik, *Structure*, 1999, **7**, 1067–1078.
- 47 P. Limphong, R. M. McKinney, N. E. Adams, B. Bennett, C. A. Makaroff, T. Gunasekera and M. W. Crowder, *Biochemistry*, 2009, **48**, 5426–5434.
- 48 S. Tottey, K. J. Waldron, S. J. Firbank, B. Reale, C. Bessant, K. Sato, T. R. Cheek, J. Gray, M. J. Banfield, C. Dennison and N. J. Robinson, *Nature*, 2008, **455**, 1138–1142.
- 49 V. N. Daitankar, S. R. Farrell and C. Thorpe, *Biochemistry*, 2009, **48**, 4828–4837.
- 50 S. E. Wiley, A. N. Murphy, S. A. Ross, P. van der Geer and J. E. Dixon, *Proc. Natl. Acad. Sci. U. S. A.*, 2007, **104**, 5318–5323.
- 51 A. R. Conlan, H. L. Axelrod, A. E. Cohen, E. C. Abresch, J. Zuris, D. Yee, R. Nechushtai, P. A. Jennings and M. L. Paddock, *J. Mol. Biol.*, 2009, **392**, 143–153.
- 52 C. H. Lillig, C. Berndt, O. Vergnolle, M. E. Lönn, C. Hudemann, E. Bill and A. Holmgren, *Proc. Natl. Acad. Sci. U. S. A.*, 2005, **102**, 8168–8163.
- 53 C. Johansson, K. L. Kavanagh, O. Gileadi and U. Oppermann, *J. Biol. Chem.*, 2007, **282**, 3077–3082.
- 54 S. Mitra and S. J. Elliott, *Biochemistry*, 2009, **48**, 3813–3815.
- 55 W. Maret, in *Handbook of Metalloproteins*, ed. A. Messerschmidt, W. Bode and M. Cygler, Wiley, Chichester, 2004, vol. 3, pp. 432–441.
- 56 F. Dardel, S. Ragusa, C. Lazennec, S. Blanquet and T. Meinel, *J. Mol. Biol.*, 1998, **280**, 501–513.
- 57 A. Becker, I. Schlichting, W. Kabsch, D. Groche, W. Schultz and A. F. Wagner, *Nat. Struct. Biol.*, 1998, **5**, 1053–1058.
- 58 K. T. Nguyen, J. C. Wu, J. A. Boylan, F. C. Gherardini and D. Pei, *Arch. Biochem. Biophys.*, 2007, **468**, 217–225.
- 59 K. W. Walker and R. A. Bradshaw, *Protein Sci.*, 1998, **7**, 2684–2687.
- 60 V. M. D'Souza and R. C. Holz, *Biochemistry*, 1999, **38**, 11079–11085.
- 61 S. C. Chai, W. L. Wang and Q. Z. Ye, *J. Biol. Chem.*, 2008, **283**, 26879–26885.
- 62 A. Serero, C. Giglione, A. Sardini, J. Martinez-Sanz and T. Meinel, *J. Biol. Chem.*, 2003, **278**, 52953–52963.
- 63 S. Escobar-Alvarez, Y. Goldgur, G. Yang, O. Ouerfelli, Y. Li and D. A. Scheinberg, *J. Mol. Biol.*, 2009, **387**, 1211–1228.
- 64 S. Fieulaine, C. Juillan-Binnard, A. Serero, F. Dardel, C. Giglione, T. Meinel and J. L. Ferrer, *J. Biol. Chem.*, 2005, **280**, 42315–42324.
- 65 X. V. Hu, X. Chen, K. C. Han, A. S. Mildvan and J. O. Liu, *Biochemistry*, 2007, **46**, 12833–12843.
- 66 J. P. Liuzzi, F. Aydemir, H. Nam, M. D. Knutson and R. J. Cousins, *Proc. Natl. Acad. Sci. U. S. A.*, 2006, **103**, 13612–13617.
- 67 T. P. Dalton, L. He, B. Wang, M. L. Miller, L. Jin, K. F. Stringer, X. Chang, C. S. Baxter and D. W. Nebert, *Proc. Natl. Acad. Sci. U. S. A.*, 2005, **102**, 3401–3406.
- 68 L. He, K. Girijashanker, T. P. Dalton, J. Reed, H. Li, H. Soleimani and D. W. Nebert, *Mol. Pharmacol.*, 2006, **70**, 171–180.
- 69 K. J. Waldron, J. C. Rutherford, D. Ford and N. J. Robinson, *Nature*, 2009, **460**, 823–830.
- 70 J.-W. Lee and J. D. Helmann, *BioMetals*, 2007, **20**, 485–499.
- 71 E. W. Althaus, C. E. Outten, K. E. Olson, H. Cao and T. V. O'Halloran, *Biochemistry*, 1999, **38**, 6559–6569.
- 72 E. Pohl, J. C. Haller, A. Mijovilovich, W. Meyer-Klaucke, E. Garman and M. L. Vasil, *Mol. Microbiol.*, 2003, **47**, 903–915.
- 73 M. A. Sheikh and G. L. Taylor, *Mol. Microbiol.*, 2009, **72**, 1208–1220.
- 74 D. Lucarelli, S. Russo, E. Garman, A. Milano, W. Meyer-Klaucke and E. Pohl, *J. Biol. Chem.*, 2007, **282**, 9914–9922.

- 75 D. A. K. Traore, A. El Ghazouani, S. Ilango, J. Dupuy, L. Jacquamet, J.-L. Ferrer, C. Caux-Thang, V. Duarte and J. M. Latour, *Mol. Microbiol.*, 2006, **61**, 1211–1219.
- 76 W. Maret and Y. Li, *Chem. Rev.*, 2009, **109**, 4682–4707.
- 77 Y. Li and W. Maret, *J. Anal. At. Spectrom.*, 2008, **23**, 1055–1062.
- 78 W. Maret, *J. Chromatogr., B: Anal. Technol. Biomed. Life Sci.*, 2009, **877**, 3378–3383.
- 79 Q. Hao, S.-H. Hong and W. Maret, *J. Cell. Physiol.*, 2007, **210**, 428–435.
- 80 A. Krezel and W. Maret, *J. Am. Chem. Soc.*, 2007, **129**, 10911–10921.
- 81 A. Krezel, Q. Hao and W. Maret, *Arch. Biochem. Biophys.*, 2007, **463**, 188–200.
- 82 H. Haase and W. Maret, *Electrophoresis*, 2008, **29**, 4169–4176.
- 83 G. Meloni, V. Sonois, T. Delaine, L. Guilloreau, A. Gillet, J. Teissié, P. Faller and M. Vasak, *Nat. Chem. Biol.*, 2008, **4**, 366–372.
- 84 G. Roesijadi, R. Bogumil, M. Vasak and J. H. R. Kagi, *J. Biol. Chem.*, 1998, **273**, 17425–17432.
- 85 U. Heinz, L. Hemmingsen, M. Kiefer and H. W. Adolph, *Chem.–Eur. J.*, 2009, **15**, 7350–7358.
- 86 W. Maret, *Biochemistry*, 2004, **43**, 3301–3309.
- 87 W. Maret, *Antioxid. Redox Signaling*, 2006, **8**, 1419–1441.
- 88 W. Maret, *Methods Enzymol.*, 2002, **348**, 230–237.
- 89 S. W. Fan, R. A. George, N. L. Haworth, L. L. Feng, J. Y. Liu and M. A. Wouters, *Protein Sci.*, 2009, **18**, 1745–1765.
- 90 W. Maret, *Methods Enzymol.*, 1993, **226**, 594–618.
- 91 Z. F. Kanyo, L. R. Scolnick, D. E. Ash and D. W. Christianson, *Nature*, 1996, **383**, 554–557.
- 92 M. Moche, J. Shanklin, A. Ghoshal and Y. Lindquist, *J. Biol. Chem.*, 2003, **278**, 25072–25080.
- 93 E. Jabri, M. B. Carr, R. P. Hausinger and P. A. Karplus, *Science*, 1995, **268**, 998–1004.
- 94 A. Volbeda and W. G. J. Hol, *J. Mol. Biol.*, 1989, **209**, 249–279.
- 95 S. K. Burley, P. R. David, R. M. Sweet, A. Taylor and W. N. Lipscomb, *J. Mol. Biol.*, 1992, **224**, 113–140.
- 96 E. Garcin, X. Vernede, E. C. Hatchikian, A. Volbeda, M. Frey and J. C. Fontecilla-Camps, *Structure*, 1999, **7**, 557–566.
- 97 C. R. Kissinger, H. E. Parge, D. R. Knighton, C. T. Lewis, L. A. Pelletier, A. Tempczyk, V. J. Kalish, K. D. Tucker, R. E. Showalter, E. W. Moomaw, L. N. Gastinel, N. Habuka, X. Chen, F. Maldonado, J. E. Barker, R. Bacquet and J. E. Villafranca, *Nature*, 1995, **378**, 641–644.
- 98 J. A. Tainer, E. D. Getzoff, K. M. Beem, J. S. Richardson and D. C. Richardson, *J. Mol. Biol.*, 1982, **160**, 181–217.
- 99 R. C. Holz, K. P. Bzymek and S. I. Swierczek, *Curr. Opin. Chem. Biol.*, 2003, **7**, 197–206.
- 100 W. T. Lowther and B. W. Matthews, *Chem. Rev.*, 2002, **102**, 4581–4608.
- 101 B. L. Vallee and D. S. Auld, *Proc. Natl. Acad. Sci. U. S. A.*, 1993, **90**, 2715–2718.
- 102 Z. Hu, T. S. Gunasekera, L. Spadafora, B. Bennett and M. W. Crowder, *Biochemistry*, 2008, **47**, 7947–7953.
- 103 T. Sugiura and Y. Noguchi, *BioMetals*, 2009, **22**, 469–477.
- 104 K. S. Larsen and D. S. Auld, *Biochemistry*, 1989, **28**, 9620–9625.
- 105 K. S. Larsen and D. S. Auld, *Biochemistry*, 1991, **30**, 2613–2618.
- 106 M. Gomez-Ortiz, F. X. Gomis-Ruth, R. Huber and F. X. Aviles, *FEBS Lett.*, 1997, **400**, 336–340.
- 107 J. T. Bukrinsky, M. J. Bjerrum and A. Kadziola, *Biochemistry*, 1998, **37**, 16555–16564.
- 108 D. S. Auld, in *Encyclopedia of Inorganic Chemistry*, ed. R. B. King, Wiley, Chichester, 2005, vol. 9, pp. 5885–5927.
- 109 F. Rusnak, L. Yu and P. Mertz, *JBIC, J. Biol. Inorg. Chem.*, 1996, **1**, 388–396.
- 110 T. A. Reiter, R. T. Abraham, M. Choi and F. Rusnak, *JBIC, J. Biol. Inorg. Chem.*, 1999, **4**, 632–644.
- 111 H. Ke and H. Wang, *Curr. Top. Med. Chem.*, 2007, **7**, 391–403.
- 112 J. A. Cowan, *JBIC, J. Biol. Inorg. Chem.*, 1997, **2**, 168–176.
- 113 J. S. Cavet, G. P. Borrelly and N. J. Robinson, *FEMS Microbiol. Rev.*, 2003, **27**, 165–181.
- 114 R. J. P. Williams, *Coord. Chem. Rev.*, 2001, **216–217**, 583–595.
- 115 C. E. Outten and T. V. O'Halloran, *Science*, 2001, **292**, 2488–2492.
- 116 R. A. Bozym, R. B. Thompson, A. K. Stoddard and C. A. Fierke, *ACS Chem. Biol.*, 2006, **1**, 103–111.
- 117 A. Krezel and W. Maret, *JBIC, J. Biol. Inorg. Chem.*, 2006, **11**, 1049–1062.
- 118 R. A. Colvin, A. I. Bush, I. Volitakis, C. P. Fontaine, D. Thomas, K. Kikuchi and W. R. Holmes, *Am. J. Physiol.: Cell Physiol.*, 2008, **294**, C726–C742.
- 119 U. Heinz, M. Kiefer, A. Tholey and H. W. Adolph, *J. Biol. Chem.*, 2004, **280**, 3197–3207.
- 120 C. A. Fierke and R. B. Thompson, *BioMetals*, 2001, **14**, 205–222.
- 121 R. Lill and U. Muhlenhoff, *Annu. Rev. Biochem.*, 2008, **77**, 669–700.
- 122 H. Shi, K. Z. Bencze, T. L. Stemmler and C. C. Philpott, *Science*, 2008, **320**, 1207–1210.
- 123 R. J. P. Williams, *J. Inorg. Biochem.*, 2002, **88**, 241–250.